

Tan Dai Ngo

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Technical Skills

- **Programming:** Python, Java, SQL, MATLAB
- **Machine Learning:** scikit-learn, XGBoost, CatBoost, regression, classification, clustering, feature engineering, Optuna
- **Data & ML Systems:** Apache Spark, Kafka, Cassandra, Neo4j, MLflow, ETL pipelines, distributed systems
- **Deployment & GenAI:** FastAPI, Docker, AWS (EC2), Sentence Transformers, Qdrant, RAG, Ollama LLM

Education

Master of Engineering in Applied Data Science | GPA: 8.68 / 9.00 *University of Victoria | Sep 2025 - Dec 2027 (Expected)*

- **Coursework:** Optimization for ML, Systems for Massive Datasets, Data Analysis & Pattern Recognition, Data Mining, Algorithms & Data Models, Artificial Intelligence

Master of Science in Computer Science

University of Chicago | Sep 2020 - Mar 2022

- **Coursework:** Algorithms, Distributed Systems, Advanced Python Programming, Databases, Applied Software Engineering

Bachelor of Science in Economics | GPA: 3.57 / 4.00

University of Washington | Sep 2015 - Aug 2019

- **Minors:** Informatics, Applied Mathematics
- **Coursework:** Applied Linear Algebra & Numerical Analysis, Econometric Theory & Practice, Elements of Statistical Methods

Experience

Math & Computer Science Teacher | APU American International Schools

Da Nang City, Vietnam | Aug 2024 - Dec 2024

- Taught Python programming, algorithms, OOP, and software engineering fundamentals to 50+ students through hands-on coding projects with focus on debugging, algorithmic reasoning, and performance optimization.

Software Developer | T-Mobile (via BeaconFire Inc.)

Bellevue, WA, USA | Jun 2022 - Apr 2024

- Automated Kafka reprocessing pipelines for 50+ weekly roaming service tests, cutting manual intervention by 80%.
- Redesigned Cassandra schemas for partner/workflow microservices (100k+ records), maintaining query latency under 3s.
- Developed Spring Boot microservices integrated with Jenkins CI/CD and Splunk, enabling scalable downstream analytics.

Projects

Distributed Ocean Data ML Platform with RAG

Systems for Massive Datasets

- Built an end-to-end distributed ML pipeline using Apache Spark (Bronze → Silver → Gold architecture) to ingest and harmonize 10M+ NOAA and ONC ocean sensor observations for scalable feature engineering and time series forecasting.
- Developed and tuned XGBoost forecasting models using Optuna and MLflow, applying chronological holdout validation on temporally correlated ocean data (RMSE: 0.00586, R²: 0.9999).
- Deployed FastAPI services for model inference and RAG search using Sentence Transformers, Qdrant, and Ollama with retrieval benchmarking (hit@k: 1.0, term recall: 0.875) and a Streamlit dashboard for interactive exploration.

End-to-End Machine Learning System for Multi-Output Fuel Blending

Shell.ai Hackathon 2025

- Built and deployed a production ML system for multi-output regression across 10 chemical blend properties using FastAPI, Docker, and AWS EC2 for real-time inference.
- Benchmarked XGBoost and CatBoost using 5-fold cross-validation, selecting CatBoost after achieving significantly lower error (MAPE: 0.64 vs. 1.29).
- Designed feature engineering and inference pipelines using weighted aggregation, mixture metrics based on entropy, and serialized preprocessing to ensure consistent production predictions.

Stock Sentiment Analysis (Team of 4)

Algorithms and Data Models

- Built a sentiment analysis pipeline over 15k+ tweets across 10 tech stocks, enabling large-scale market sentiment analysis, by integrating X API data, SQLite/Neo4j storage, and FinTwitBERT modeling.
- Identified measurable sentiment-return relationships, including statistically significant predictive effects for NVIDIA, by implementing engagement-weighted sentiment aggregation and Granger causality testing.
- Analyzed equity volatility dynamics across major tech stocks, revealing persistent volatility clustering, by applying GARCH(1,1) modeling and automated cross-stock analysis workflows.